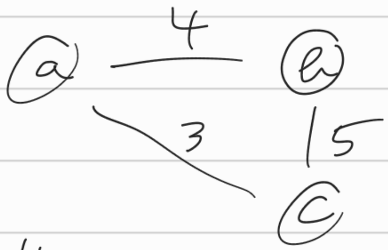
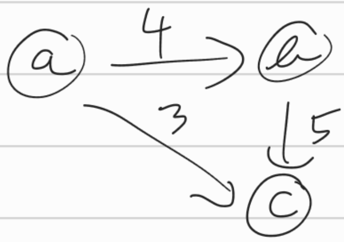
9. óra

Minimális feszítőfa

Grafikus, szöveges ábrázolás:

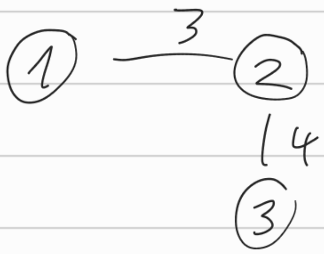


a − b, 4; c, 3
b − c, 5

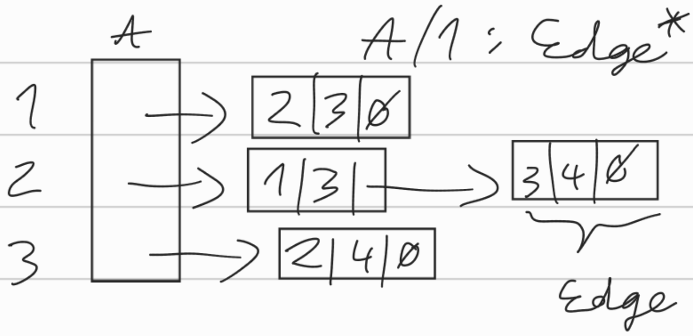


a → b, 4; c, 3
b → c, 5

Éllistás ábrázolás:

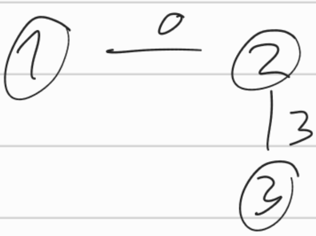


A/1 : Edge* [n]



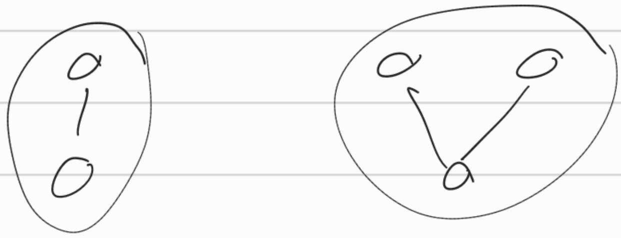
Edge
+ v : ℕ
+ w : ℝ
+ next : Edge*

Csúcsmátrixos ábrázolás:

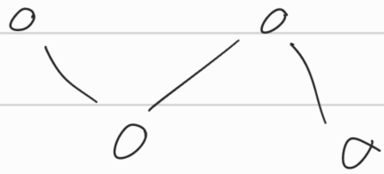


	1	2	3
1	0	0	∞
2	0	0	3
3	∞	3	0

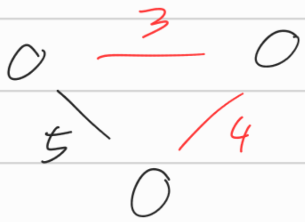
– feszítőerdő:



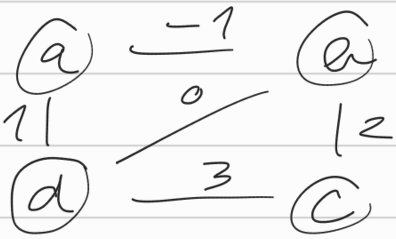
– feszítőfa:



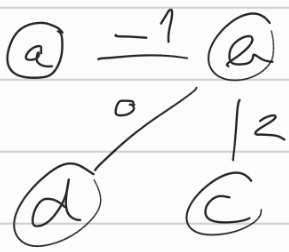
– minimális feszítőfa:



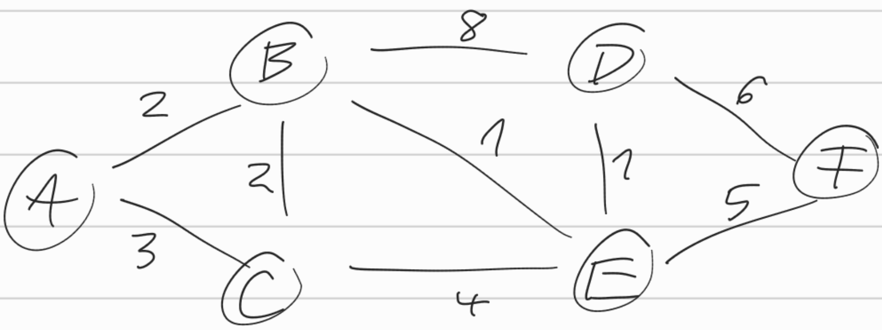
Kruskal algoritmusa:



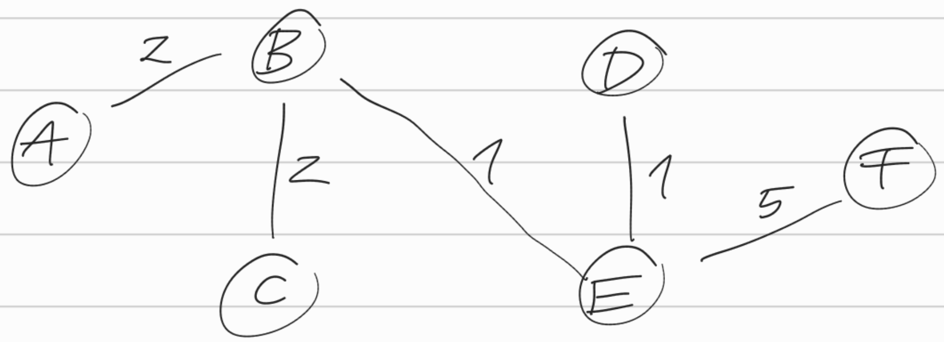
komponensek (fák)	él	súly	biztonságos-e?
a, b, c, d	a – b	−1	+
ab, c, d	b – d	0	+
abd, c	a – d	1	–
abd, c	b – c	2	+
abcd			



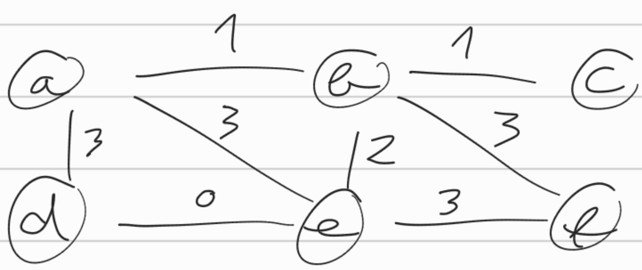
Feladat:

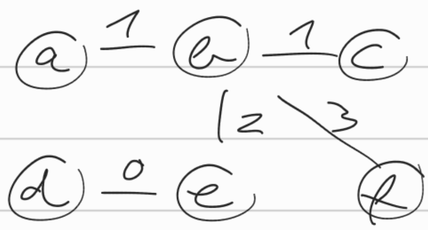


komponensek	él	súly	biztonságos-e?
A, B, C, D, E, F	B – E	1	+
A, BE, C, D, F	D – E	1	+
A, BDE, C, F	B – C	2	+
A, BCDE, F	A – B	2	+
ABCDE, F	A – C	3	–
ABCDE, F	C – E	4	–
ABCDE, F	E – F	5	+
ABCDEF			



Feladat:

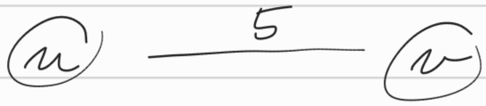




komponensek	él	súly	biztonságos-e?
a, b, c, d, e, f	d – e	0	+
a, b, c, de, f	a – b	1	+
ab, c, de, f	b – c	1	+
abc, de, f	b – e	2	+
abcde, f	a – d	3	–
abcde, f	a – e	3	–
abcde, f	b – f	3	+
abcdef			

Prim algoritmusa:

- p: szülő csúcs
- c: adott csúcs és a szülője közti él súlya



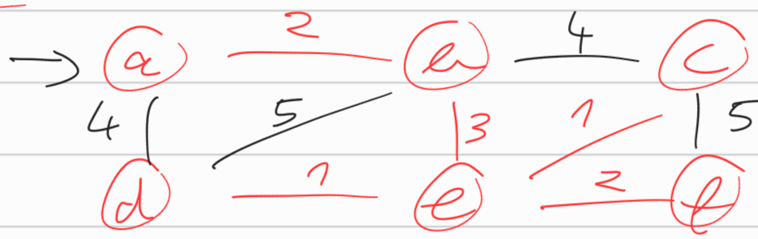
$p(u) = v$
$c(u) = 5$

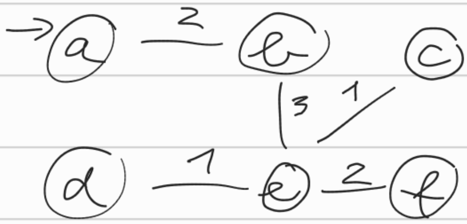
$p(r) = \emptyset$
$c(r) = 0$

$p(l) = \emptyset$
$c(l) = \infty$

l-t még nem vettük hozzá az MST-hez

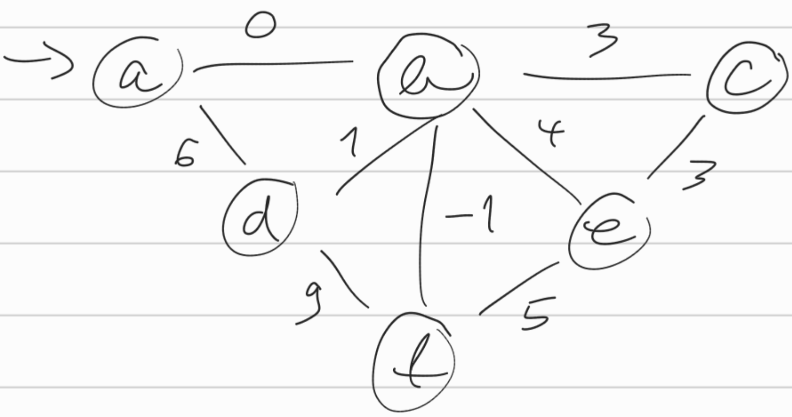
Feladat:





c: a	b	c	d	e	f	mst	p: a	b	c	d	e	f
$\boxed{0}$	∞	∞	∞	∞	∞	—	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
	$\boxed{2}$	∞	4	∞	∞	a		a		a		
		4	4	$\boxed{3}$	∞	b			b		b	
		$\boxed{1}$	1		2	e			e	e		e
			$\boxed{1}$		2	c						
					$\boxed{2}$	d						
						f						
0	2	1	1	3	2		$\emptyset$	a	e	e	b	e

Feladat:



c						mst	π					
a	b	c	d	e	f		a	b	c	d	e	f
$\boxed{0}$	∞	∞	∞	∞	∞	—	∅	∅	∅	∅	∅	∅
	$\boxed{0}$	∞	6	∞	∞	a		a		a		
		3	1	4	$\boxed{-1}$	b			b	b	b	b
		3	$\boxed{1}$	4		f						
		$\boxed{3}$		4		d						
				$\boxed{3}$		c					c	
						e						
0	0	3	1	3	-1		∅	a	b	b	c	b